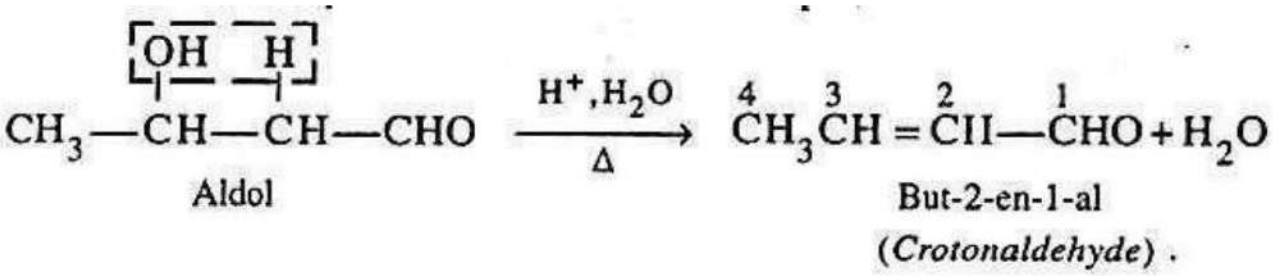
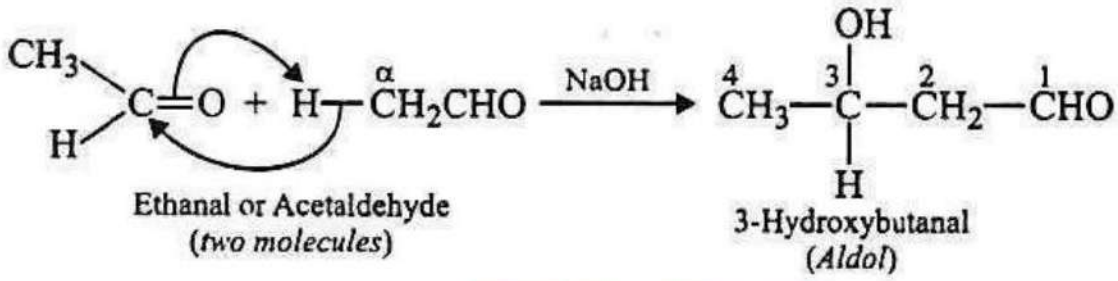
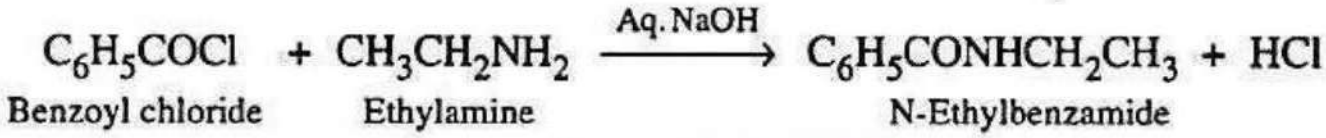
44. Aldol Condensation



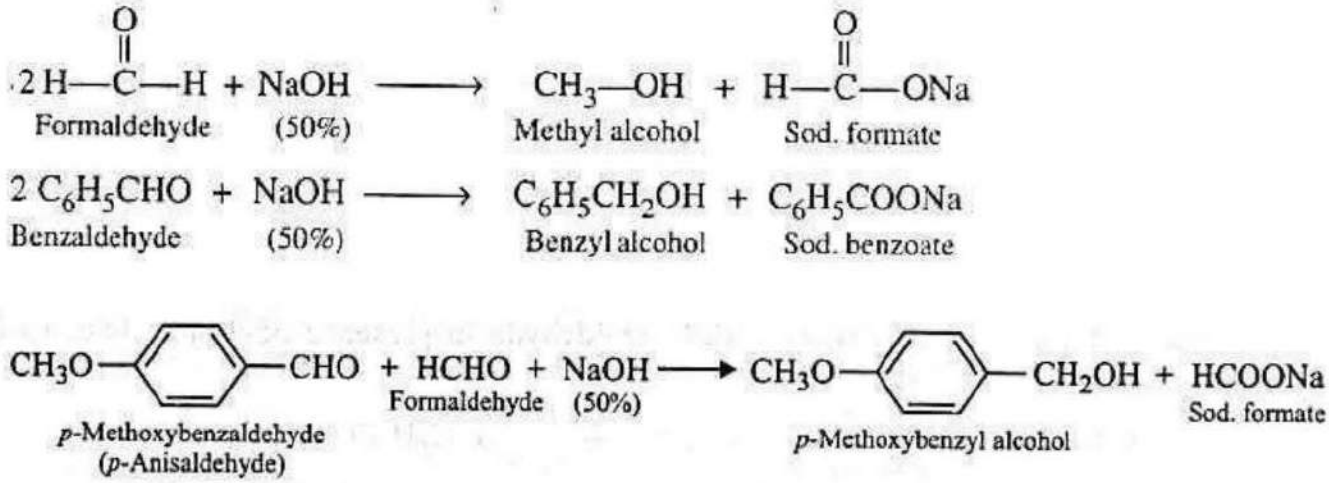
45. Aldol Reaction



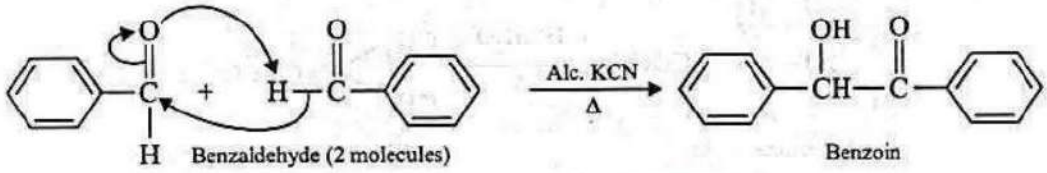
46. Benzoylation Reaction



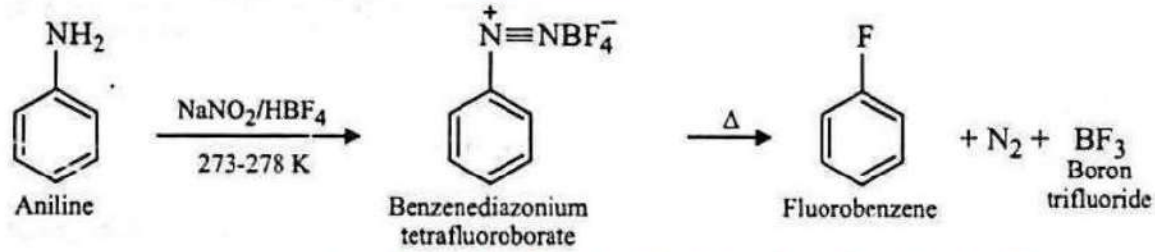
40. Cannizzaro Reaction



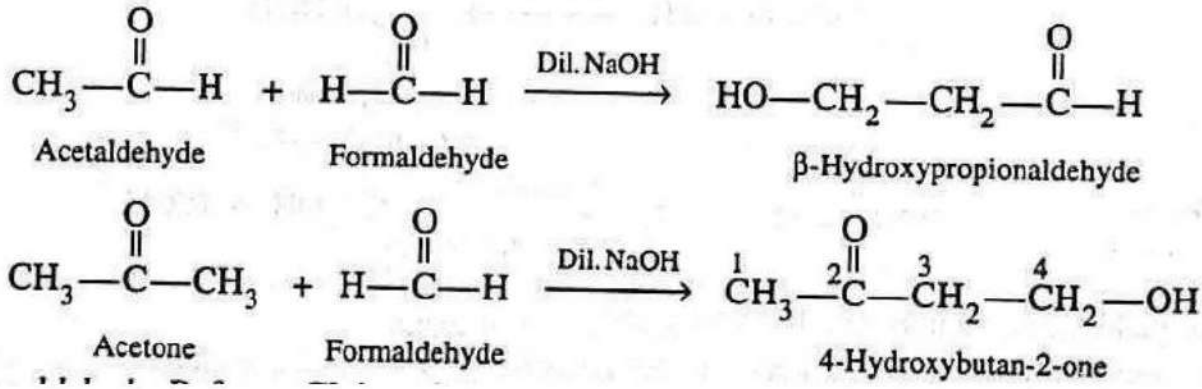
41. Benzoin Condensation



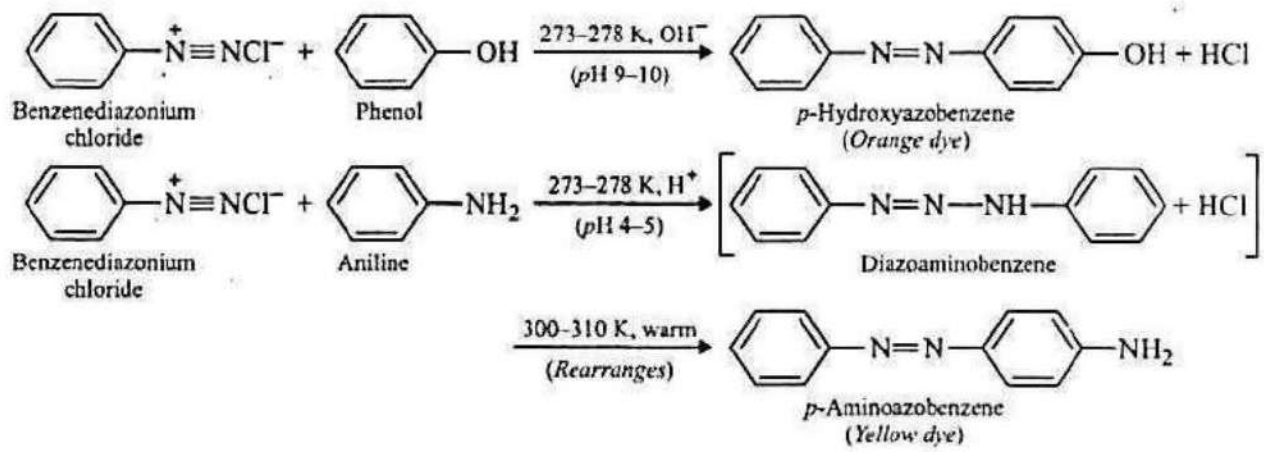
42. Balg-Schiemann Reaction



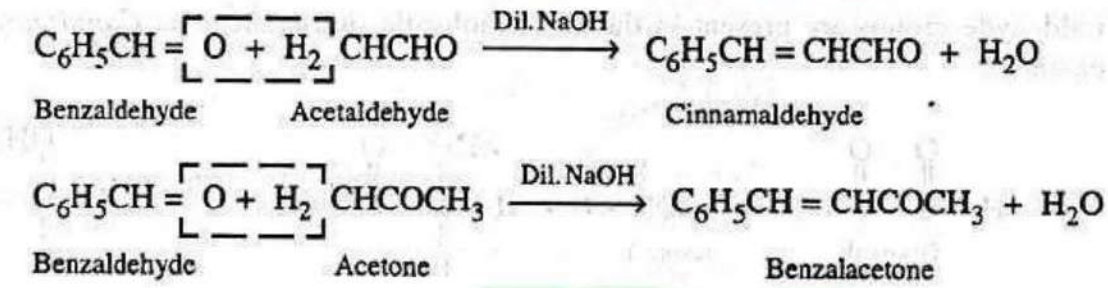
43. Cross Aldol Condensation



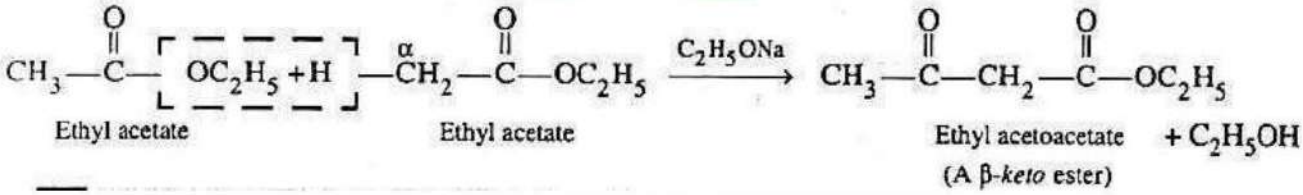
36. Coupling Reaction



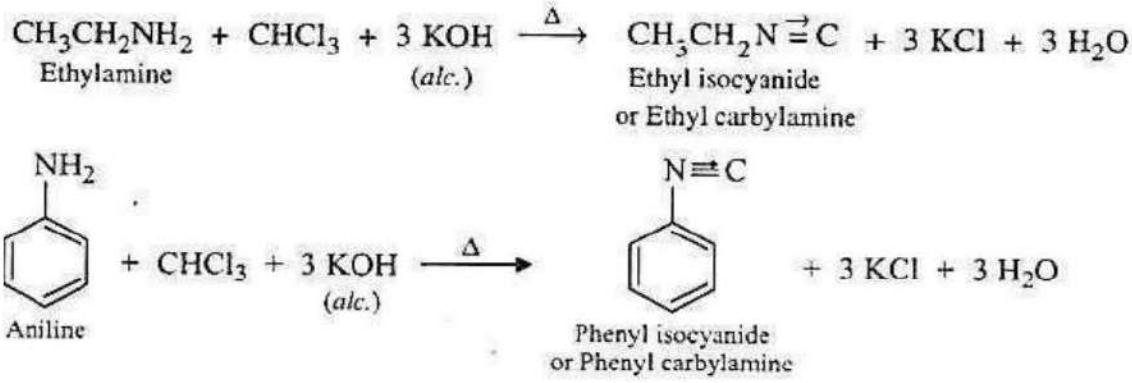
37. Claisen Reaction



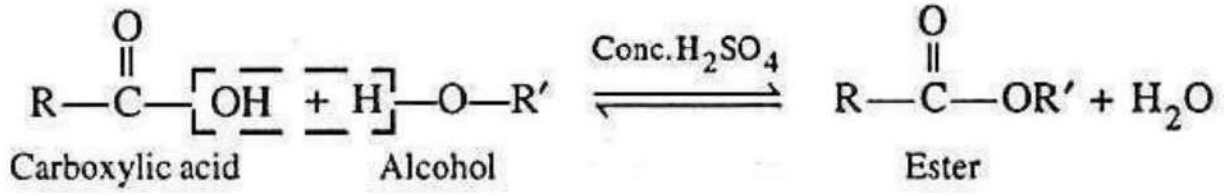
38. Claisen Condensation



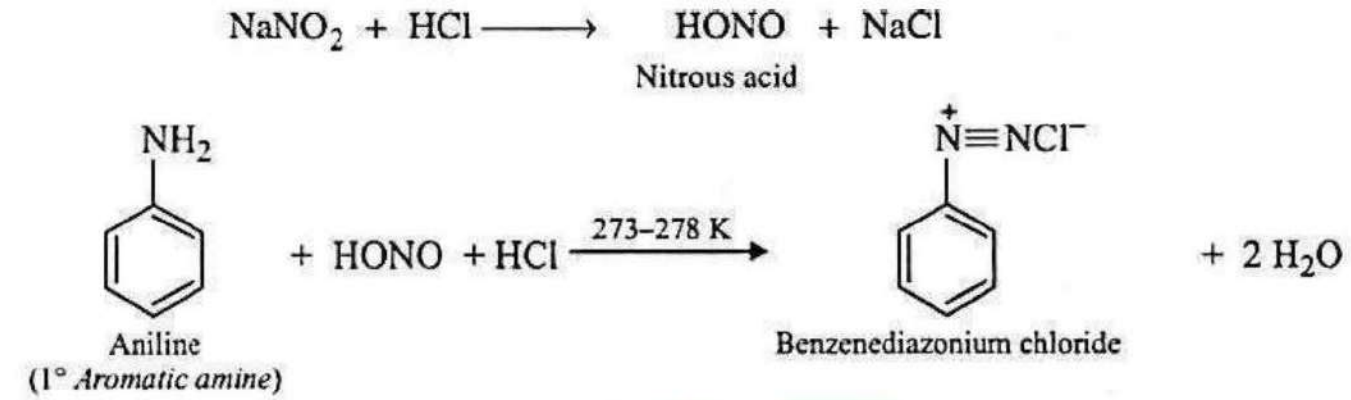
39. Carbylamine Reaction



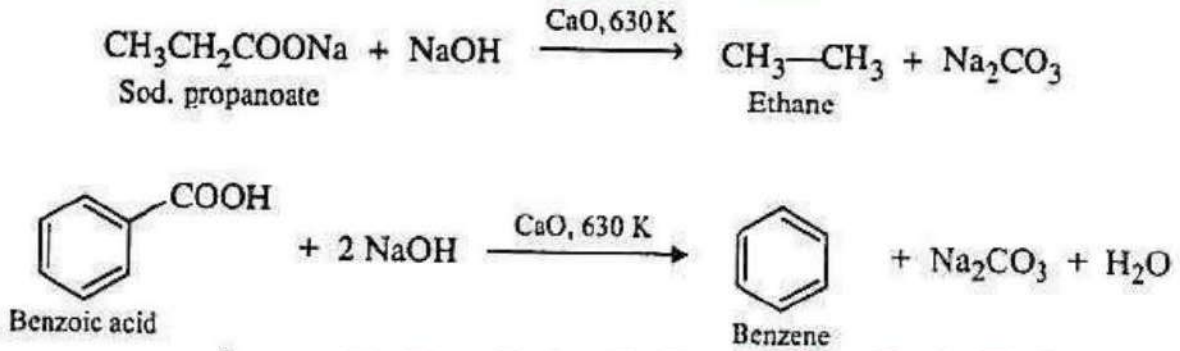
32. Esterification Reaction



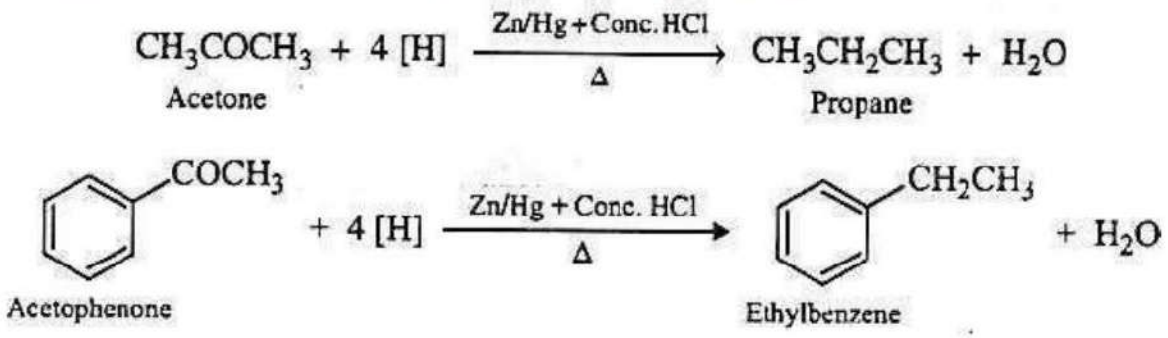
33. Diazotization Reaction



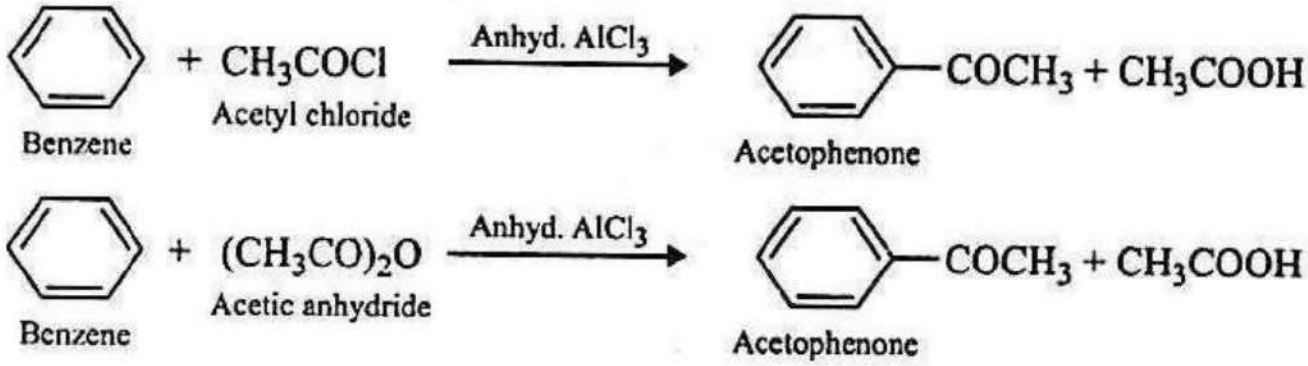
34. Decarboxylation Reaction



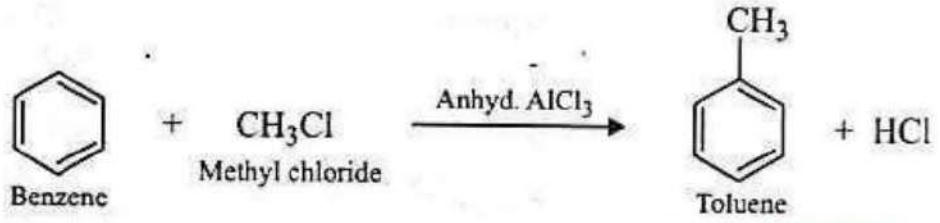
35. Clamensen Reduction



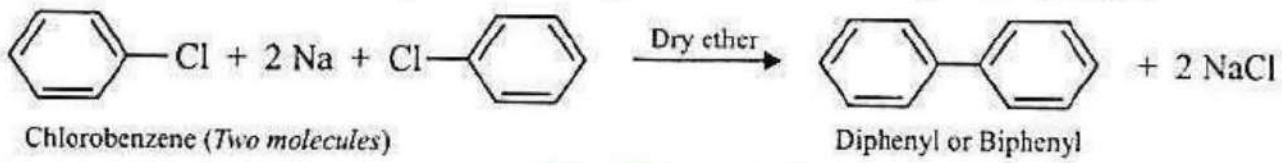
27. Friedel-CraftsAcylation



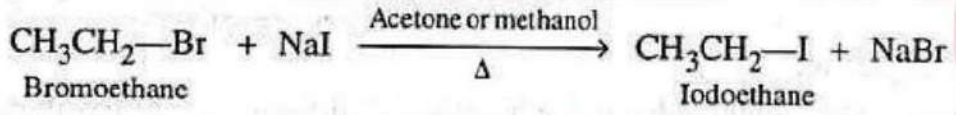
28. Wurtz-Fittig Reaction



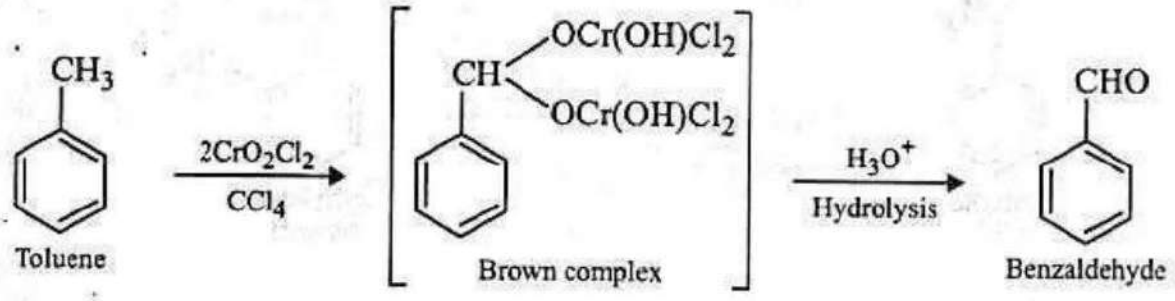
29. Fittig Reaction



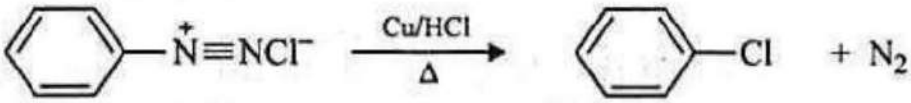
30. Finkelstein Reaction



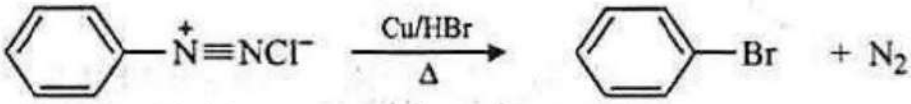
31. Etard Reaction



24. Gatterman Reaction

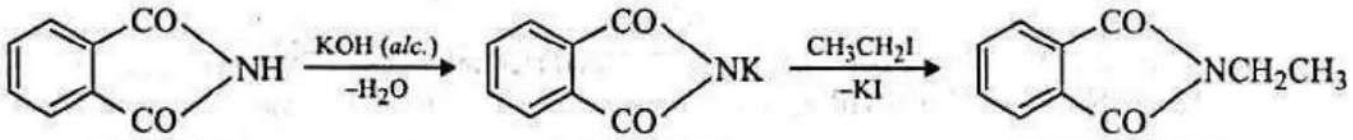


Benzenediazonium chloride Chlorobenzene

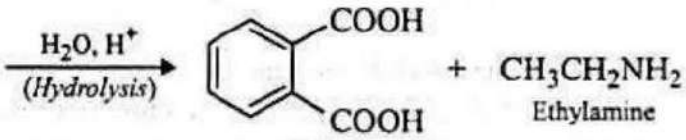


Benzenediazonium chloride Bromobenzene

25. Gabriel Phthalimide Reaction

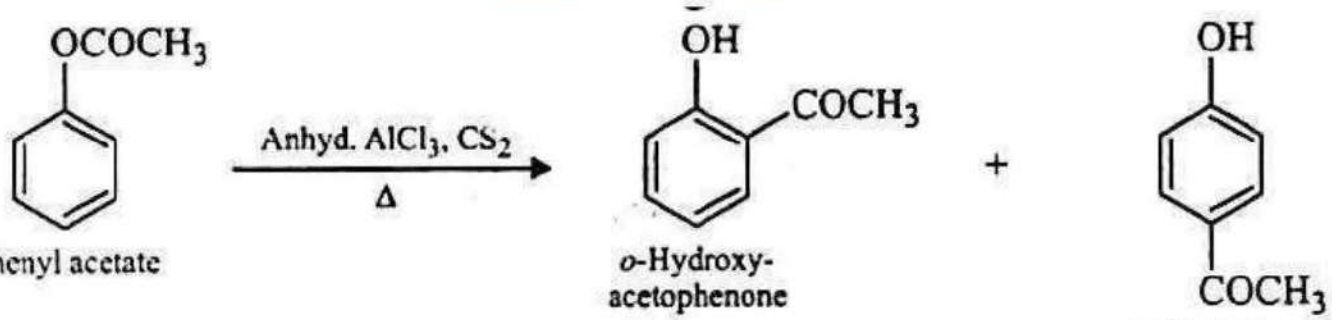


Phthalimide Pot. phthalimide N-Ethylphthalimide



Phthalic acid Ethylamine

26. Fries rearrangement



Phenyl acetate *o*-Hydroxy-acetophenone *p*-Hydroxy-acetophenone

19. Hofmann bromamide Reaction

$$\underset{\text{Acetamide}}{CH_3CONH_2} + Br_2 + 4\,NaOH \xrightarrow{\Delta} \underset{\text{Methylamine}}{CH_3NH_2} + 2\,NaBr + Na_2CO_3 + 2\,H_2O$$

20. Holmann Ammonolysis

$$(i)\quad \underset{\text{Ammonia}}{H_3N:} + \underset{\text{Alkyl halide}}{R-X} \xrightarrow{373\,K} \underset{\text{1° Amine salt}}{R\overset{+}{N}H_3X^-} \overset{NH_3}{\rightleftharpoons} \underset{\text{1° Amine}}{R-NH_2} + NH_4^+X^-$$

21. Hell-Volhard Zelinsky Reaction

$$\underset{\text{Acetic acid}}{\overset{\alpha}{C}H_3COOH} \xrightarrow[-HCl]{Cl_2,P} \underset{\text{Monochloro-acetic acid}}{Cl\overset{\alpha}{C}H_2COOH} \xrightarrow[-HCl]{Cl_2,P} \underset{\text{Dichloro-acetic acid}}{Cl_2\overset{\alpha}{C}HCOOH} \xrightarrow[-HCl]{Cl_2,P} \underset{\text{Trichloro-acetic acid}}{Cl_3CCOOH}$$

22. Iodoform Reaction

$$\underset{\text{Ethanol}}{CH_3CH_2OH} + 4\,I_2 + 6\,NaOH \longrightarrow \underset{\text{Iodoform}}{CHI_3} + HCOONa + 5\,NaI + 5\,H_2O$$

$$\underset{\text{2-Propanol}}{CH_3CHOHCH_3} + 4\,I_2 + 6\,NaOH \longrightarrow \underset{\text{Iodoform}}{CHI_3} + CH_3COONa + 5\,NaI + 5\,H_2O$$

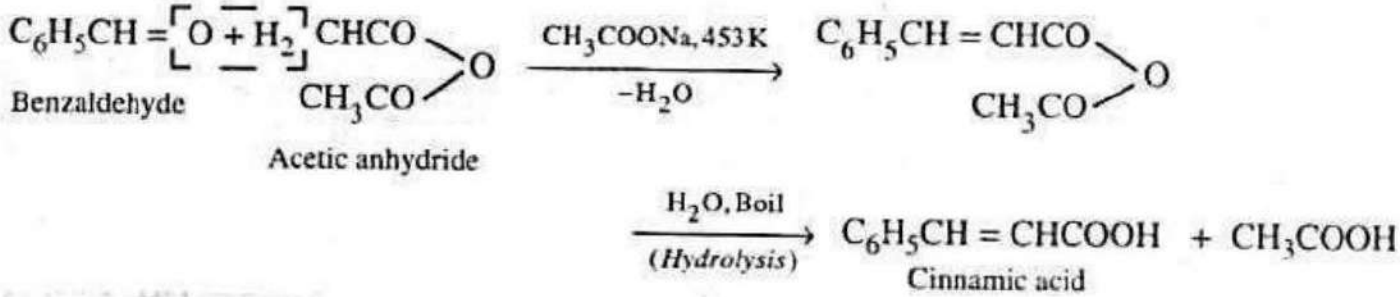
$$\underset{\text{Propanone}}{CH_3COCH_3} + 3\,I_2 + 4\,NaOH \longrightarrow \underset{\text{Iodoform}}{CHI_3} + CH_3COONa + 3\,NaI + 3\,H_2O$$

23. Gattermann-Koch Reaction

$$CO + HCl \longrightarrow \underset{\text{Formyl chloride (}\textit{unstable}\text{)}}{HCOCl}$$

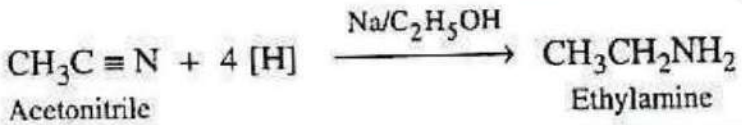
$$C_6H_6 + HCOCl \xrightarrow{AlCl_3+CuCl} \underset{\text{Benzaldehyde}}{C_6H_5CHO} + HCl$$

14. Perkin Reaction



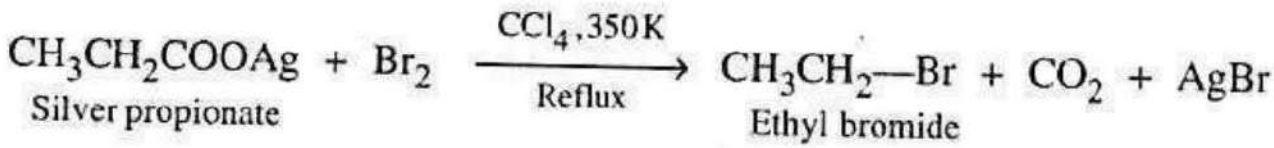
15. Liebermann's nitroso Reaction

16. Mendius Reaction

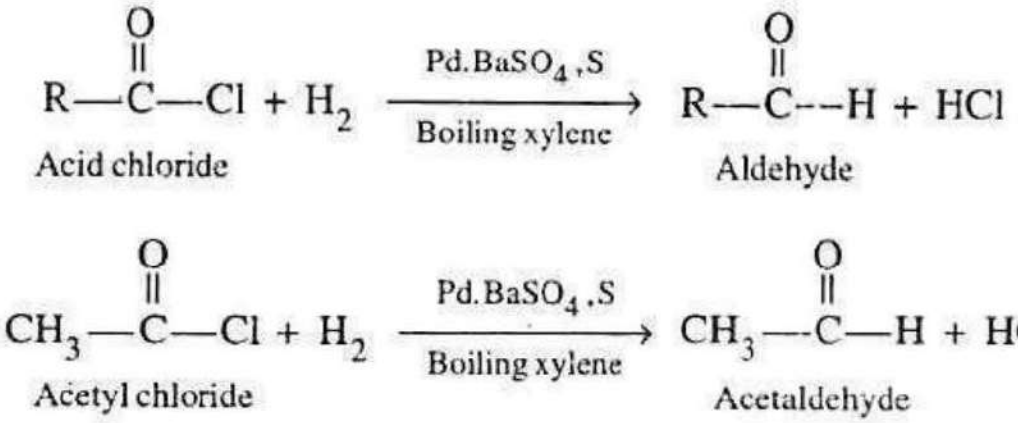


17. Colbe-Schmitt Reaction

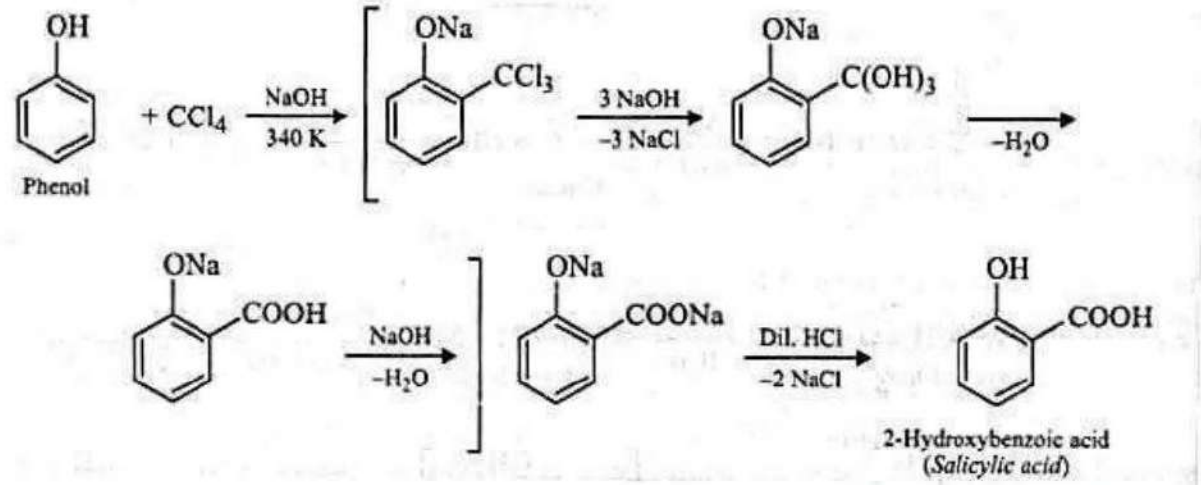
18. Hunsdicker Reaction



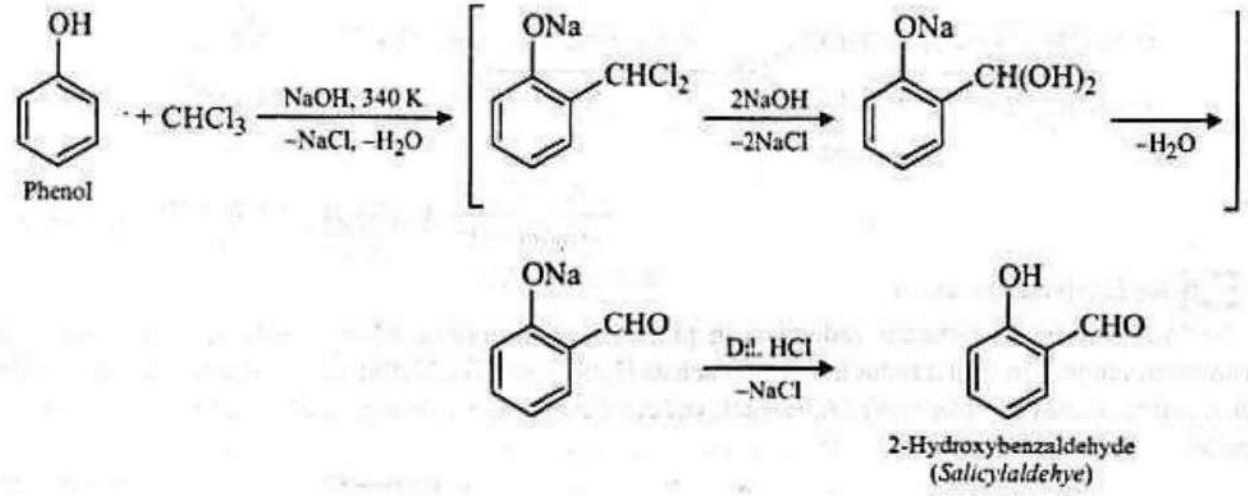
11. Rosenmund Reaction



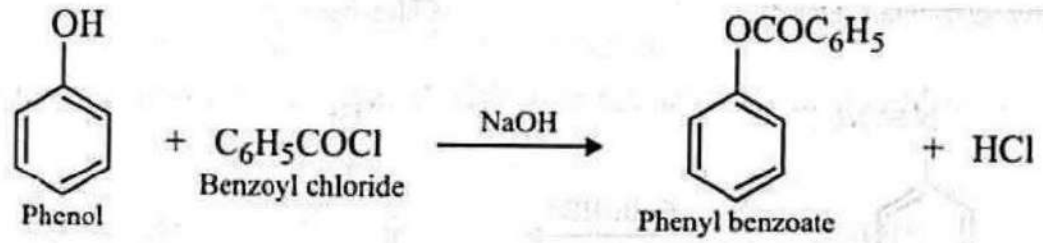
12. Reimer-TiemannReaction

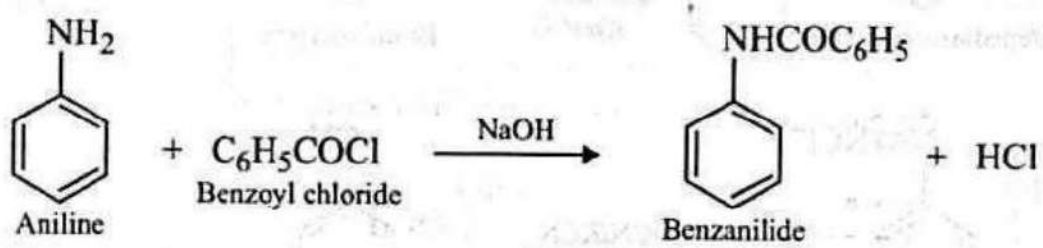


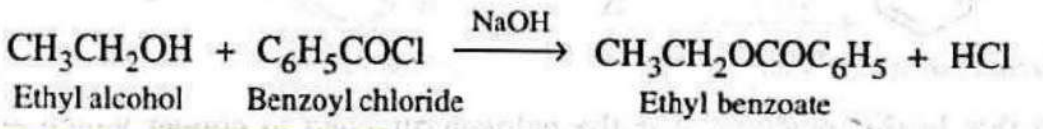
13. Reimer-TiemannReaction



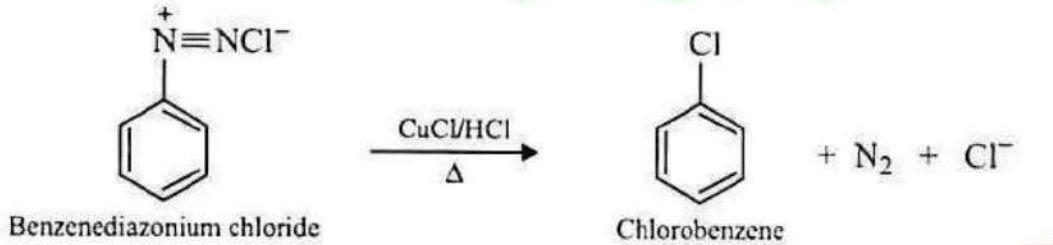
9. Schotten Baumann Reaction

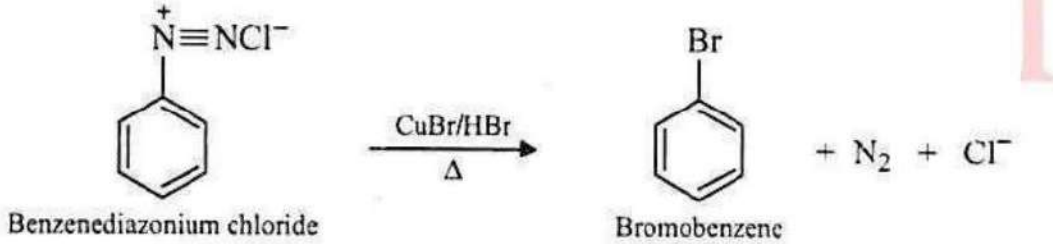


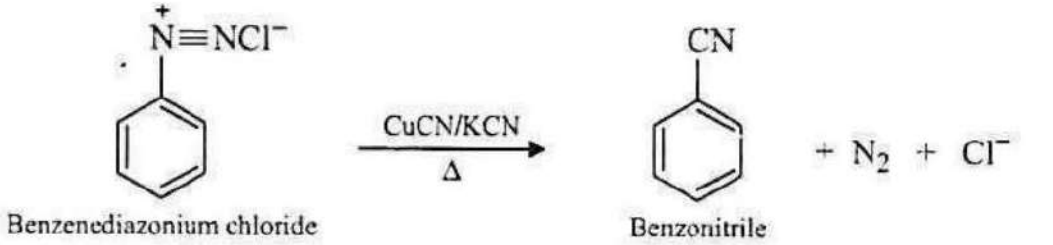




10. Sandmeyer's Reaction

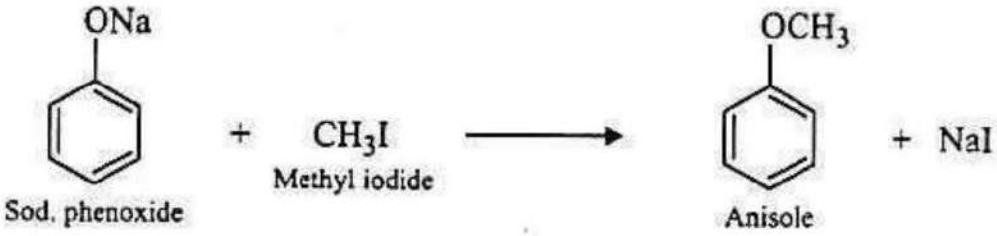






5. Wiliamsons Reaction

$$\underset{\text{Ethyl iodide}}{CH_3CH_2I} + \underset{\text{Sod. ethoxide}}{CH_3CH_2ONa} \longrightarrow \underset{\text{Diethyl ether}}{CH_3CH_2{-}O{-}CH_2CH_3} + NaI$$



6. StephenReaction

$$SnCl_2 + 2\,HCl \longrightarrow SnCl_4 + 2\,[H]$$

$$\underset{\text{Acetonitrile}}{CH_3C{\cdot}{\equiv}N} + 2\,[H] + HCl \xrightarrow[290-295\,K]{\text{Dry ether}} \underset{\text{Acetaldimine hydrochloride}}{CH_3CH = NH.HCl}$$

$$\downarrow \text{Boiling } H_2O$$

$$\underset{\text{Acetaldehyde}}{CH_3CH = O} + NH_4Cl$$

7. SwartsReaction

$$\underset{\text{Bromomethane}}{CH_3Br} + AgF \xrightarrow{\Delta} \underset{\text{Fluoromethane}}{CH_3F} + AgBr$$

$$\underset{\text{Chloroethane}}{2\,CH_3CH_2Cl} + Hg_2F_2 \xrightarrow{\Delta} \underset{\text{Fluoroethane}}{2\,CH_3CH_2F} + Hg_2Cl_2$$

8. SchmidtReaction

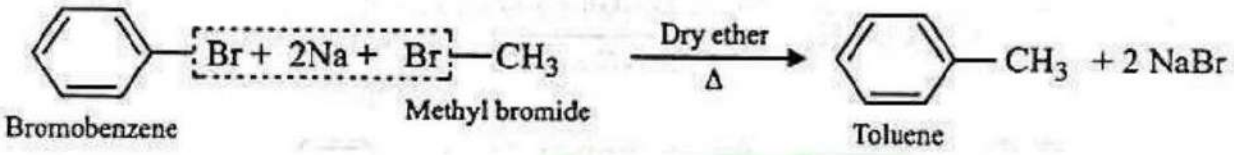
$$\underset{\text{Benzoic acid}}{C_6H_5COOH} \xrightarrow[+\text{conc.}H_2SO_4]{N_3H} \underset{\text{Aniline}}{C_6H_5{-}NH_2} + CO_2 + N_2$$

Name Reactions (Organic Chemistry)

1. Acetylation

$$\underset{\text{Benzoyl chloride}}{C_6H_5COCl} + \underset{\text{Ethyl alcohol}}{HOCH_2CH_3} \xrightarrow{\text{Aq. NaOH}} \underset{\text{Ethyl benzoate}}{C_6H_5COOCH_2CH_3} + HCl$$

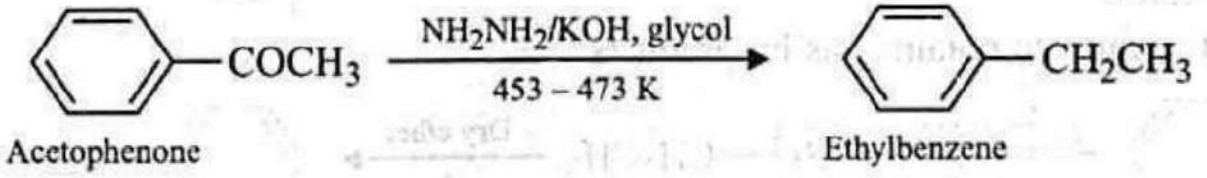
2. Wurtz FittigReaction



3. WurtzReaction

$$\underset{\text{Methyl bromide}}{CH_3-Br} + 2Na + \underset{\text{Methyl bromide}}{Br-CH_3} \xrightarrow{\text{Dry ether}} \underset{\text{Ethane}}{CH_3-CH_3} + 2\,NaBr$$

4. Wolff-KishnerReduction

$$\underset{\text{Aldehyde}}{R-CH=O} \xrightarrow[-H_2O]{NH_2NH_2} \underset{\text{Hydrazone}}{[R-CH=NNH_2]} \xrightarrow[453-473\,K]{KOH,\ glycol} R-CH_3 + N_2$$

$$\underset{\text{Acetone}}{CH_3COCH_3} \xrightarrow[453-473\,K]{NH_2NH_2/KOH,\ glycol} \underset{\text{Propane}}{CH_3CH_2CH_3}$$